References

Textbook: Intro. to algorithms

Thomas Cormen
Charles Leiserson
Ronald Rivest
Clifford Stein

Wikipedia.

- Examples of algo.
- Runtime analysis of algo:
  insertion sort
- Big-Oh notation

Definition:

Computational procedure that takes an input and produce an output that achieve a certain goal.

Eg: Sorting problem

Input: $\{31, 41, 59, 26, 41, 58\}$

Output: $\{26, 31, 41, 41, 58, 59\}$

input: $\langle a_1, \cdots, a_n \rangle$

output: $\langle a_1', a_2', \cdots, a_n' \rangle$

such that it is a reording of $\langle a_1 \cdots a_n \rangle$ in increasing order.

Def: Instance of problem

Def: Correctness of an algo.

Eg: Substring matching problem.

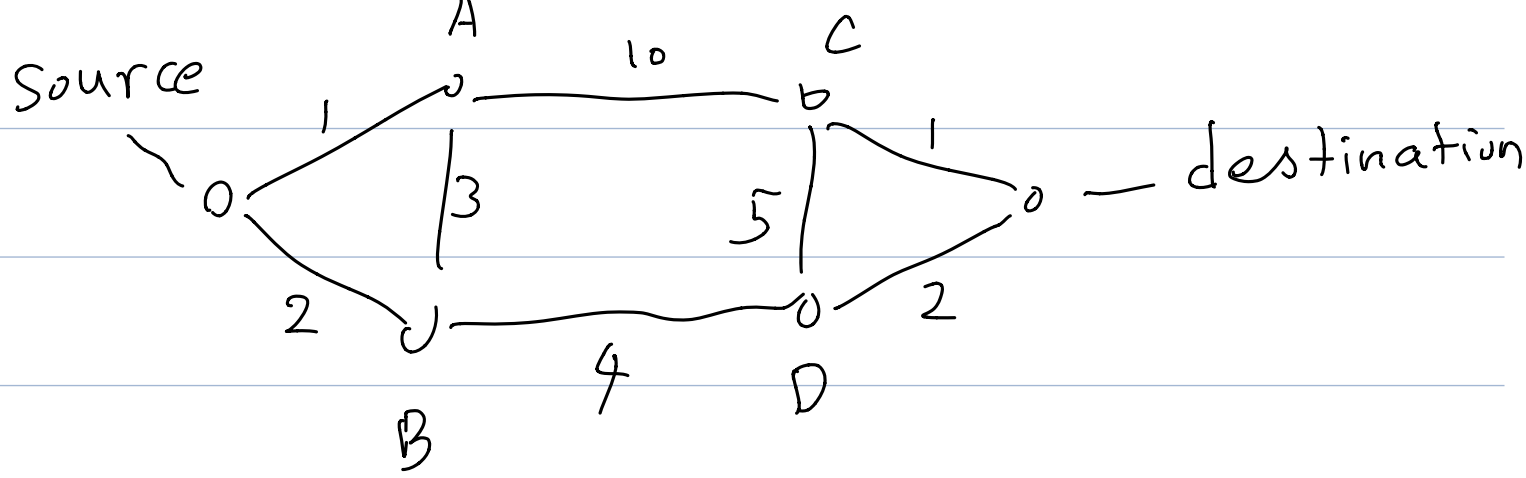
Input.

$$\begin{cases} \text{Text: } ATG\underline{CCG}TA \\ \text{Pattern: } \underline{CCG} \end{cases}$$

Output : 4

Eg: Shortest path problem.

Input : A graph



Output: Source — B — D — Dest.

A path . . — — — — —

<u>Eg:</u> Longest common substring

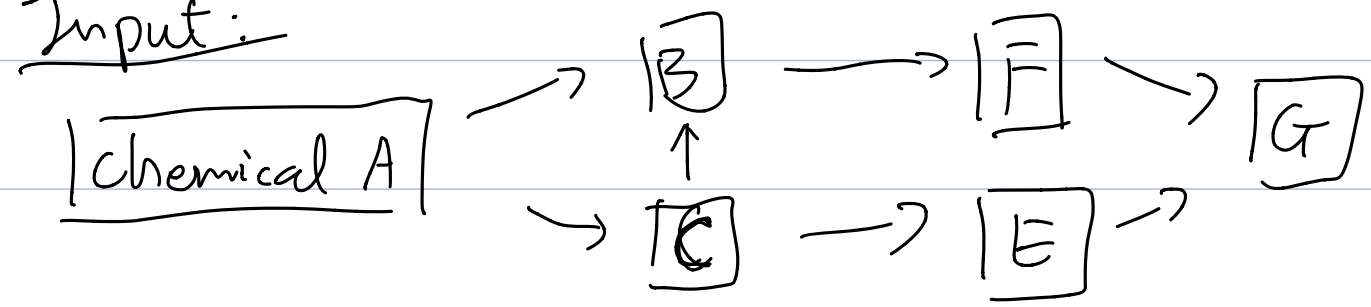
Input: $\begin{cases} \text{ABCDEFG} \\ \text{AEF2GBCDE} \end{cases}$

Output : BCDE

<u>Hint</u>: Dynamic programming.

Eg: Topological sort problem.

Input:

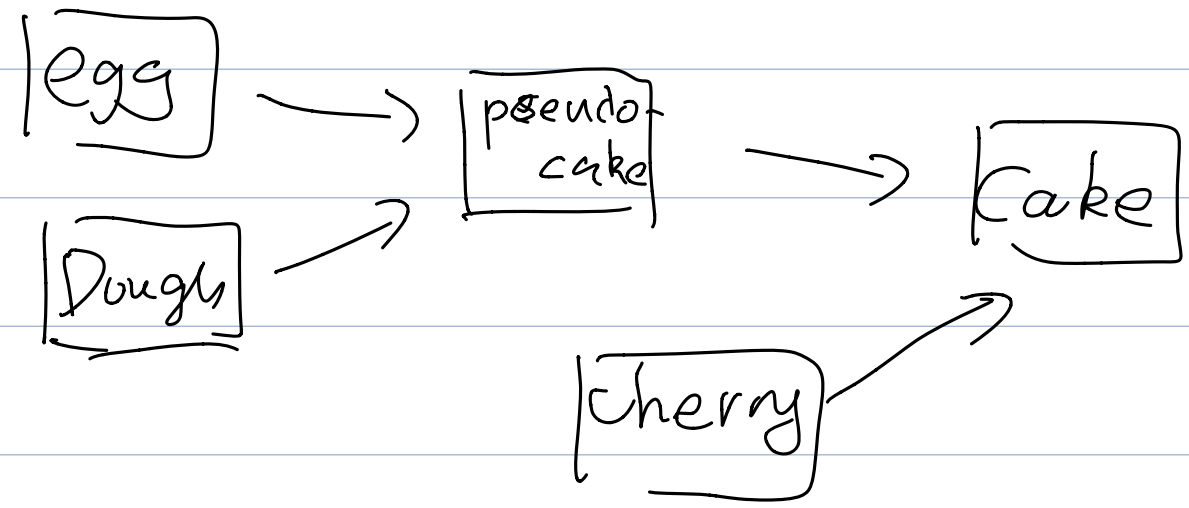


Output: an ordering such that predessesors appear first.

$A \to C \to B \to F \to E \to G$

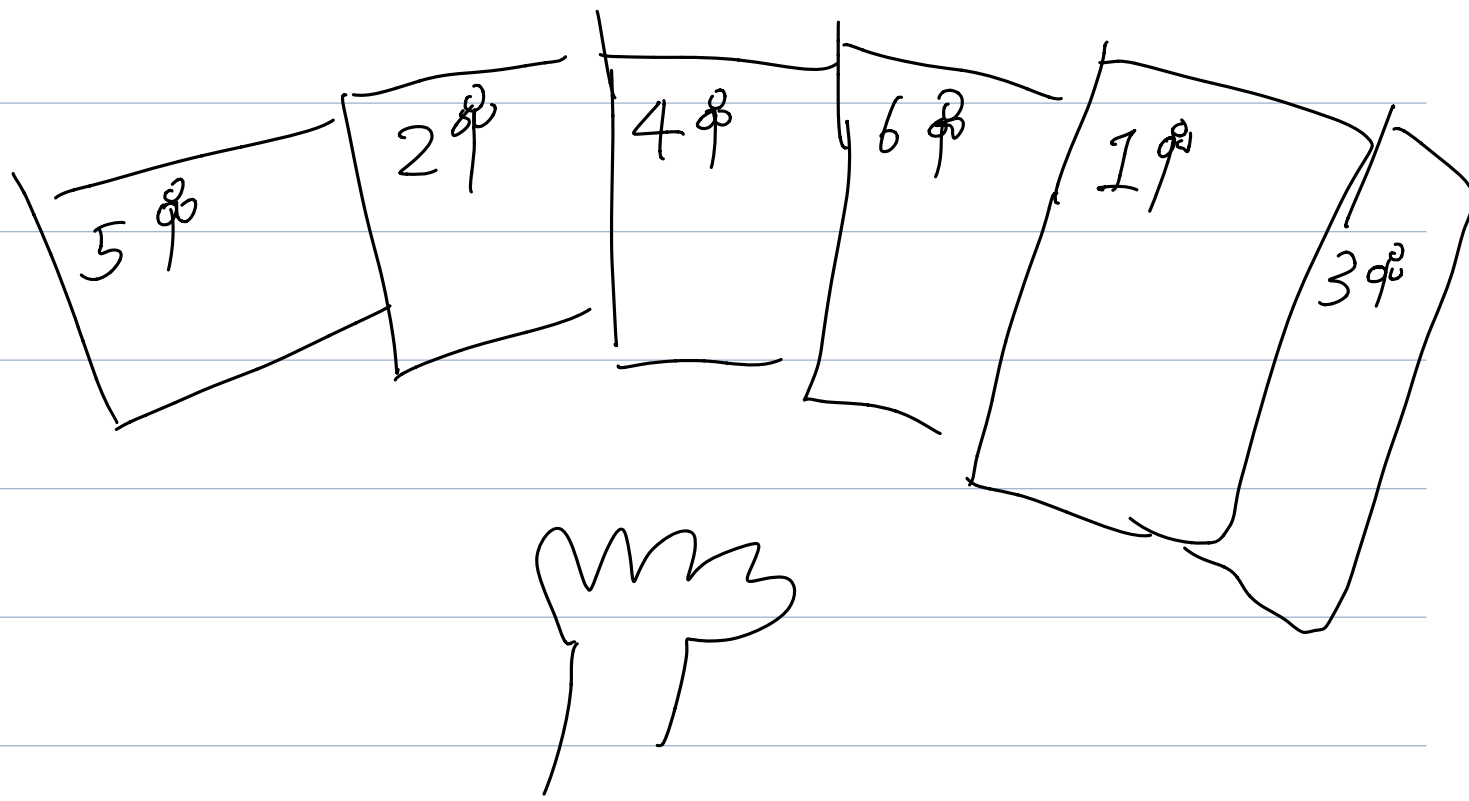
$\to E \to F$

Another analogy



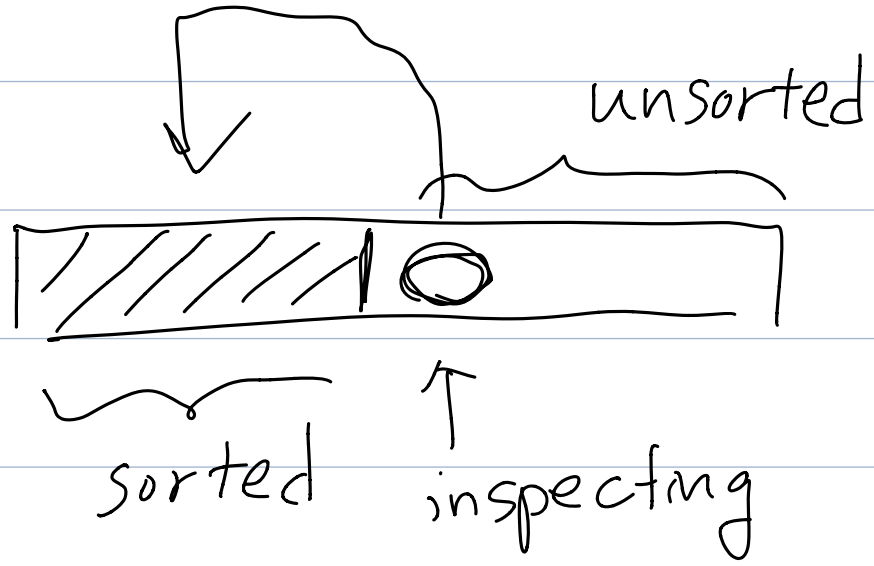
Part II : Run-time analysis of algorithm : insertion sort.

Eg : Input $\{5, 2, 4, 6, 1, 3\}$



1. 2, 4, 5, 6, 1, 3

2. 1, 2, 4, 5, 6, 3



insert numbers to the sorted subarray.

1. 5 | 2, 4 6, 1, 3

sorted inspect

2. 2 5 | 4 6 1 3

sorted inspect

3. 2 4 5 | 6 1 3

sorted

4. 2 4 5 6 | 1 3

5. 1 2 4 5 6 | 3

6. 1 2 3 4 5 6

Quiz

$\boxed{1, 2, 3, 4, 5}$

increasing order

step 1: ' ' ' ' ' ' '

step 2: ' ' - - - - ..

<u>Def</u>: In-place sorting: Everything done in the input array

<u>Def</u>: Input size.

<u>Def</u>: Running time:

= how many "steps" are needed.

1 add

2. compare

3. change number in an array.

Def: Worst-case analysis.

T_A : instance $\to \mathbb{R}^+$ (time)

time needed to finish when instance is received

$$\max_{\text{instance}} T_A(\text{instance}) \geq$$

$\{1, \cdots n\}$ ← length-n arrays.

Def: Average-case analysis.

$$\sum T_A(\text{instance}) \cdot \text{prob. instance occurs}$$

$\boxed{\text{Insertion sort.}}$

Best-case : $O(n)$

Worst-case :

increasing order

$\boxed{5 \mid 4 \;\; 3 \;\; 2 \;\; 1}$ $\Rightarrow$ 1 comp.

2 access $\longleftrightarrow$ change

$\boxed{4 \;\; 5 \mid 3 \;\; 2 \;\; 1}$ $\Rightarrow$ 2 comp.

3 access

$\boxed{3 \;\; 4 \;\; 5 \mid 2 \;\; 1}$ $\Rightarrow$ 3

4

$\boxed{2 \;\; 3 \;\; 4 \;\; 5 \mid 1}$ $\Rightarrow$ 4

5

$$\text{Time} = \underbrace{(1+2+3+4)}_{\text{comparisons}} + \underbrace{(2+\cdots+5 \;\; - - -)}_{\text{accesses}}$$

n-length

$$(1+2+\cdots+(n-1)) = \frac{n(n+1)}{2}$$

$$= O(n^2)$$

<u>Q</u>: $1+2^2+\cdots+n^2 = O(n^3)$

$$1^a + 2^a + \cdots + n^a = O(n^{a+1})$$

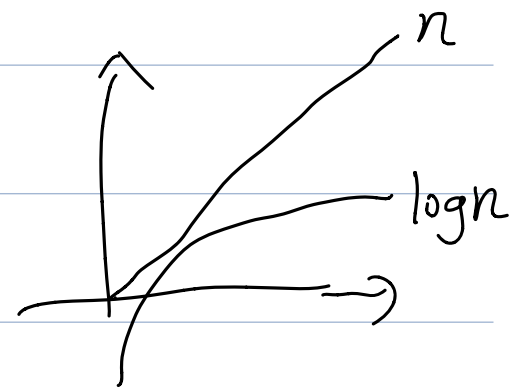
a integer

$\boxed{\text{Average-case}}$ $O(n^2)$

Q: Is it best ?

Average case

$O(n^2)$ — Insertion, Bubble, selection



$O(n\log n)$ — Merge, Quick, Heapsort

$O(n)$-type — Counting, Radix, Bucket (need more than just comparison)

quicker

$a_1 = 1 \quad 2 \quad 3 \quad 4 \quad 5$

$a_2 = 2 \quad 3 \quad 4 \quad 5 \quad 6$

<u>Rule of thumb of big-Oh notation</u>

$$2n^2 + 1n + 2 = O(n^2)$$

$$n^3 + 2n^2 + 1 = O(n^3)$$

<u>Reference</u> : Chapter II. 8

Sorting in linear time